

		Increases, drag force increases.
12	C	When X falls from height h, loss in G.P.E. becomes K.E. at the bottom. $\frac{1}{2}mv^2 = mgh$ When X collides with Y, applying conservation of momentum, $mv = 2mv_1$ $v_1 = \frac{v}{2}$ Applying conservation of energy, $\frac{1}{2}(2m)v_1^2 = (2m)gt$ $\frac{1}{2}(2m)\left(\frac{v}{2}\right)^2 = (2m)gt$ $t = \frac{h}{4}$
13	D	The centripetal force on the satellite of mass m is $\sqrt{\frac{GMm}{r}}$